METRIC

FF-B-171/17
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, PRIMARILY RADIAL, COUNTERBORED OUTER RING, CONTACT ANGLE UP TO AND INCLUDING 10°, NON-SEPARABLE, ANSI/AFBMA DIMENSION SERIES 02.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

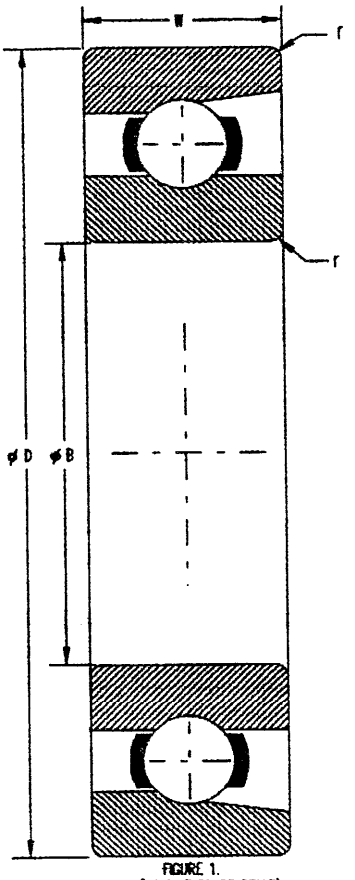


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:45Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size code in table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

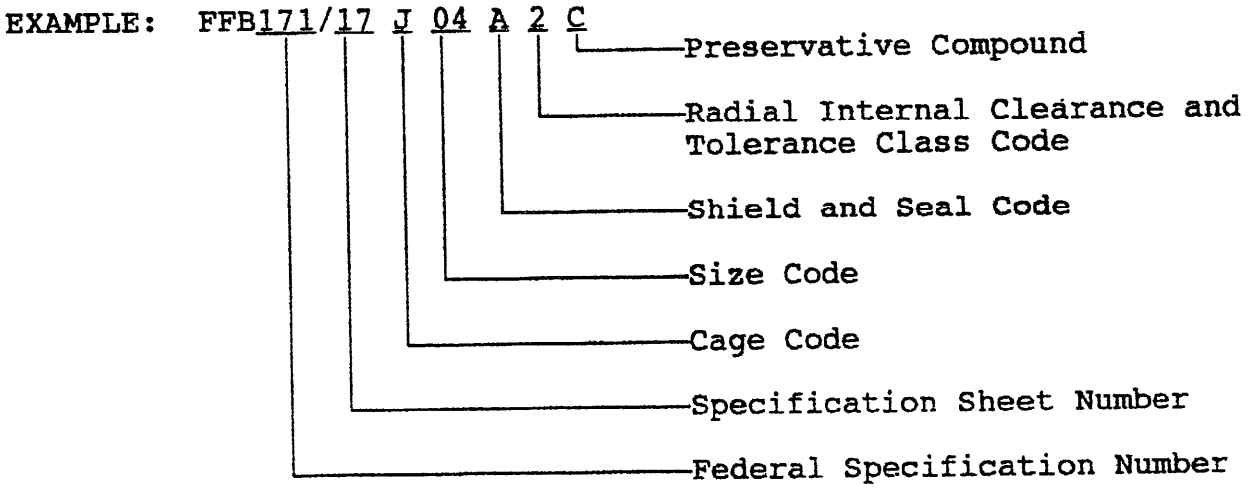
3. Shields and Seals. All primarily radial bearings with counterbored outer rings shall be open (no shields or seals, see figure 1) and designated by the code "A" in the part number.

4. Radial Interval Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

5. Tolerance Class . The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

6. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and preservative compound.



FFB171/17J04A2C indicates - Bearing, cage-pressed steel or molded plastic; bore 20mm; outside diameter 47mm; width 14mm; open (no shields or seals); radial internal clearance. symbol 0 (normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:45Z
Check the source to verify that this is the current version before use.

TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
00	10	30	9	0.6	15	1190
01	12	32	10	0.6	17	1350
02	15	35	11	0.6	20	1620
03	17	40	12	0.6	22	2005
04	20	47	14	1.0	26	2315
05	25	52	15	1.0	31	2800
06	30	62	16	1.0	36	3100
07	35	72	17	1.1	42	4315
08	40	80	18	1.1	47	5700
09	45	85	19	1.1	52	5680
10	50	90	20	1.1	57	6170
11	55	100	21	1.5	63.5	8960
12	60	110	22	1.5	68.5	9750
13	65	120	23	1.5	73.5	11800
14	70	125	24	1.5	78.5	12715
15	75	130	25	1.5	83.5	14580
16	80	140	26	2.0	90	13925
17	85	150	28	2.0	95	16990
18	90	160	30	2.0	100	21665
19	95	170	32	2.1	107	22480
20	100	180	34	2.1	112	25480
21	105	190	36	2.1	117	27505
22	110	200	38	2.1	122	29960
24	120	215	40	2.1	132	32415
26	130	230	40	3.0	144	36365
28	140	250	42	3.0	154	38985
30	150	270	45	3.0	164	44625
32	160	290	48	3.0	174	50470
34	170	310	52	4.0	188	54270
36	180	320	52	4.0	198	56350
38	190	340	55	4.0	208	61610
40	200	360	58	4.0	218	64715
42	210	380	61	3.0	232	71390
44	220	400	65	4.0	238	75315
46	230	420	68	3.0	254	82075

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:45Z
Check the source to verify that this is the current version before use.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
- AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-17)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:45Z
Check the source to verify that this is the current version before use.